

PRANEETH REDDY

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Data Engineer with 5+ years of experience designing, architecting, and delivering scalable **cloud-native data platforms**, **enterprise data warehouses**, **data lakes**, and distributed data processing solutions across retail, banking, financial services, and insurance domains. Strong expertise in developing end-to-end **ETL/ELT pipelines**, real-time and batch data processing frameworks, and analytics-ready data products using **Python**, **SQL**, **Apache Spark**, **PySpark**, **AWS**, **Snowflake**, **Airflow**, **Kafka**, and modern cloud technologies. Experienced in integrating large-scale structured, semi-structured, and streaming data from diverse enterprise systems, APIs, databases, and third-party sources while implementing robust **data quality**, **data governance**, data lineage, security, and monitoring frameworks to ensure accuracy, reliability, and compliance. Proven track record leading **cloud migration** and **data modernization** initiatives, optimizing distributed processing workloads, improving pipeline performance and scalability, and building high-performance **data architectures** that support **enterprise analytics**, **business intelligence**, **machine learning**, regulatory reporting, and data-driven decision-making. Adept at collaborating with cross-functional teams including data scientists, analysts, software engineers, product managers, and business stakeholders to deliver secure, reliable, and **production-grade data solutions** in fast-paced, highly regulated environments.

WORK EXPERIENCE

Data Engineer | Kroger | Irving, TX | Aug 2024 - Present

- Architected and maintained enterprise-scale **ETL/ELT pipelines** using **Python**, **SQL**, and **Apache Spark**, processing billions of customers, transactional, and supply chain records supporting merchandising, loyalty, and operational analytics initiatives.
- Designed and implemented **cloud-native data solutions** leveraging **AWS S3**, **Redshift**, **Glue**, and **BigQuery**, enabling scalable ingestion, transformation, and delivery of analytics-ready datasets across multiple business domains.
- Built metadata-driven **data ingestion frameworks** integrating structured and semi-structured data from 20+ internal and external sources, reducing manual onboarding effort by 40%.
- Optimized **Spark workloads**, partitioning strategies, and distributed SQL processing, improving pipeline performance by 35% while enhancing scalability and resource utilization.
- Implemented automated **data quality controls**, reconciliation frameworks, monitoring, and observability solutions that improved data reliability and reduced production data incidents.
- Developed **dimensional data models** and curated analytics-ready datasets supporting executive reporting, customer analytics, forecasting, and machine learning use cases.
- Led **cloud migration initiatives** transitioning legacy on-premise workloads to modern cloud architectures while maintaining business continuity and regulatory compliance.
- Established **CI/CD deployment frameworks** using Git, Docker, and automated testing processes, improving deployment reliability and reducing release cycle times.
- Partnered with product, analytics, and engineering teams to define enterprise data requirements and deliver scalable **data products** supporting strategic business initiatives.
- Mentored junior engineers through code reviews, technical documentation, engineering standards, and platform governance best practices.

Data Engineer | 7-Eleven | Mumbai, India | Jan 2021 - Jul 2023

- Designed and developed scalable **Spark-based ETL pipelines** processing multi-million-record customer, transactional, inventory, and operational datasets supporting enterprise reporting and analytics platforms.
- Built enterprise **data integration solutions** across APIs, flat files, relational databases, and third-party systems, enabling reliable ingestion into centralized **data warehouse** environments.
- Implemented **Change Data Capture (CDC)** and **SCD Type 2** frameworks supporting historical tracking, audit readiness, and regulatory reporting requirements.
- Engineered distributed **Apache Spark** workloads reducing processing times by 40% while improving platform scalability for growing data volumes.

- Developed **dimensional data models** and analytics-ready datasets supporting customer analytics, operational reporting, forecasting, and executive dashboards.
- Automated **data validation**, reconciliation, and anomaly detection frameworks using Python and SQL, improving data quality and reducing manual validation efforts by 35%.
- Optimized complex SQL workloads across Oracle, Hive, and SQL Server environments, improving query performance by 60% and enhancing reporting responsiveness.
- Collaborated with analysts, data scientists, and business stakeholders to translate business requirements into scalable, production-grade data solutions.
- Maintained **data lineage documentation**, governance artifacts, operational runbooks, and monitoring processes supporting enterprise data management standards.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, PySpark, R
- **Data Engineering & Architecture:** ETL/ELT Development, Data Pipeline Design, Data Integration, Data Modeling, Star Schema, Snowflake Schema, Data Warehousing, Data Lakes, Lakehouse Architecture, Change Data Capture (CDC), Slowly Changing Dimensions (SCD Type 2), Metadata Management, Data Lineage, Master Data Management (MDM), Data Governance
- **Big Data & Distributed Processing:** Apache Spark, PySpark, Kafka, Hive, Hadoop, Distributed Data Processing, Batch Processing, Real-Time Data Processing
- **Cloud Platforms:** AWS (S3, Redshift, Glue, Lambda, IAM, CloudWatch), Azure Data Factory, Azure Storage, GCP BigQuery, Cloud Data Migration, Cloud-Native Architecture
- **Databases & Data Warehousing:** Snowflake, Redshift, BigQuery, PostgreSQL, MySQL, SQL Server, Oracle, DB2, Query Optimization, Performance Tuning, Database Design
- **Orchestration & DevOps:** Apache Airflow, Git, Docker, CI/CD Pipelines, Automated Deployments, Monitoring & Observability, Logging Frameworks, MLflow
- **Data Quality & Governance:** Data Validation, Data Reconciliation, Data Profiling, Data Quality Frameworks, Root Cause Analysis, Audit Readiness, Regulatory Compliance, Data Security Best Practices
- **Analytics & BI:** Tableau, Power BI, KPI Reporting, Dashboard Development, Self-Service Analytics
- **Methodologies:** Agile, Scrum, SDLC, Technical Documentation

PROJECT

Enterprise Data Lake Modernization Platform | AWS, Apache Spark, Airflow, Snowflake, Python, SQL

- Designed and implemented a scalable **cloud-native data lake platform** that consolidated structured and semi-structured data from 15+ enterprise systems into a centralized analytics environment using **AWS S3, Apache Spark, Airflow**, and **Snowflake**.
- Developed automated **ETL/ELT pipelines** processing 500M+ records monthly, incorporating data validation, reconciliation, lineage tracking, and monitoring frameworks that improved data reliability and reduced manual intervention by 40%.
- Built dimensional data models and analytics-ready datasets supporting enterprise reporting, customer analytics, operational dashboards, and machine learning workloads, reducing data delivery latency by 35% and improving stakeholder access to trusted data assets.

EDUCATION

Master of Science in Data Science | University of Texas at Arlington | Arlington, TX | GPA: 4.0/4.0

Relevant Coursework: Database Systems, Big Data Analytics, Data Mining, Machine Learning, Statistical Modeling, Data Visualization, Cloud Computing, Data Warehousing, Distributed Computing, Applied Data Science